25901114

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Skill Enhancement Course (SEC) - MG4SECBTG200 - QUALITY CONTROL IN BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A-One word or sentence-Answer any 10 questions

Each question carries 2 marks

SCHEME

1. Name any principle of quality assurance. [K] / [CO1]
   Customer focus or continuous improvement/process approach)- 2 marks
2. CAPA in audit stands for? [A] / [CO3]
   Corrective and preventive action- 2 marks
3. Identify an advantage of Food Safety Modernisation Act. [U] / [CO4]
   Focus on prevention of food safety hazards – 2 marks
4. Recall the law passed in India to prevent deceptive food from reaching the public.[K] / [CO4]
   Food Safety and Standards Act (FSSA), 2006 – 2 marks
5. Name one key component of GLP. [K] / [CO5]
   Documentation (SOP-Standard operating procedures) - 2 marks
6. State the discipline that applies control charts in quality control. [K] / [CO2]
   Statistical Quality Control (SQC) - 2 marks
7. WHO stands for? [K] / [CO3]

World Health Organization – 2 marks

8. State a benefit of FDA. [K] / [CO4]

   Ensures safety and efficacy of drugs and food products- 2 marks

9. What is used to sterilize equipments? [U] / [CO3]

   Autoclave/ Steam sterilization- 2 marks

10. Identify the QC aspect that prevents data manipulation. [U] / [CO2]

    Data integrity (ALCOA principles)- 2 marks

11. Name one aim of Food Safety. [U] / [CO4]

    To ensure food is safe for consumption- 2 marks

12. Mention one example of a liquid chemical sterilant. [U] / [CO3]

    Gluteraldehyde (Hydrogen peroxide)/Anyone - 2 marks

[2x10 = 20]

Part B-Short Essay-Answer any 5 questions

Each question carries 4 marks

13. Explain QC in Electrophoresis. [U] / [CO3]

    Definition- 1 mark, Marker, instrument calibration, documentation- 3 marks

14. Differentiate the role of RDAC and DCGI in India. [U] / [CO5]

    Difference- 4 points- 4 marks

15. Discuss about histogram in quality control. [U] / [CO3]

    Frequency graph- 1mark, identifies distribution- 2 marks, helps decision making- 1mark

16. Explain the importance of documentation and record-keeping in Quality Assurance and Quality Control systems.[U] / [CO2]

    Any documentation in QC- 4 marks

17. Describe the best practices for QC in ELISA and Western Blot. [U] / [CO3]

    ELISA – 4 points- 2marks, Western Blot- 4 points- 2 marks.

18. How can personal hygiene help in HACCP? [K] / [CO4]

    Hygiene in HACCP- 4 marks

19. Explain Data Integrity in Current Good Manufacturing Practices (CGMP). [K] / [CO5]

    ALCOA principle(4 points)- 4 marks [4x5 = 20]

Part C-Essay type questions-Answer any 1 questions

Each question carries 10 marks

20. How can be prepare control chart for data analysis? [A] / [CO3]

    Steps(Select→collect→ calculate→control→plot→analyze→Act)- 10 marks

21. Evaluate the role of documentation and records in GMP. [U] / [CO5]

    Structure- definition, types, uses, examples- 10 marks

[10x1 = 10]